



RIVER VALLEY HIGH SCHOOL
H2 CHEMISTRY

Suggested Solutions for 2015 H2 Chemistry A Level Papers

Paper 1

1	B	11	D	21	B	31	D
2	B	12	C	22	D	32	A
3	C	13	D	23	B	33	C
4	C	14	B	24	A	34	A
5	B	15	D	25	D	35	B
6	D	16	D	26	D	36	A
7	B	17	A	27	D	37	C
8	C	18	C	28	C	38	C
9	A	19	B	29	B	39	A
10	B	20	A	30	A	40	B

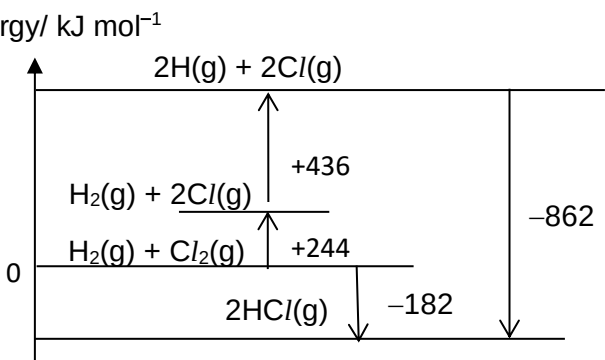
Paper 2

1	(a)	CH₃CH₂CO₂H + CH₃OH ⇌ + H₂O	[1]
	(b)	<u>(A) Determination of accurate concentration of HCl</u> 1. Pipette 10.0 cm³ of approximately 3 mol dm⁻³ HCl using a 10.0 cm³ pipette into a 250 cm³ conical flask. 2. Add 1-2 drops of phenolphthalein indicator to the solution. 3. Fill a 50.0 cm³ burette with 1.00 mol dm⁻³ NaOH. Titrate the HCl in the conical flask & observe the end point when the colour changes from colourless to pale pink, adding dropwise near the end point. 4. Repeat titration until consistent titre values (V₁) within ± 0.10 cm³ are obtained. 5. Calculation of accurate concentration $\frac{V_1}{1000} \times 1.00 = n_{HCl}$ $[HCl] = \frac{n_{HCl}}{\frac{10.0}{1000}} \text{ (mol dm}^{-3}\text{)}$	[11]

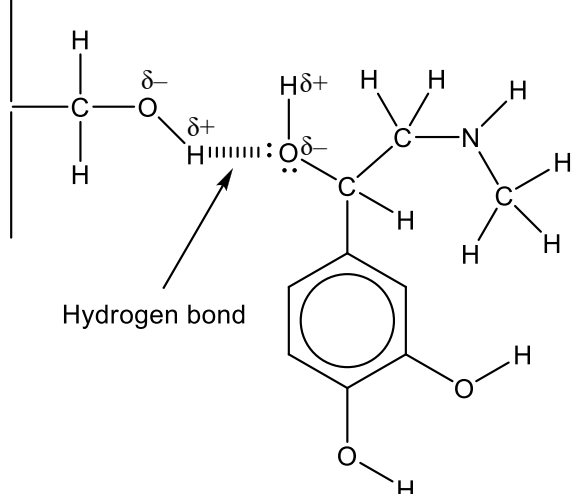
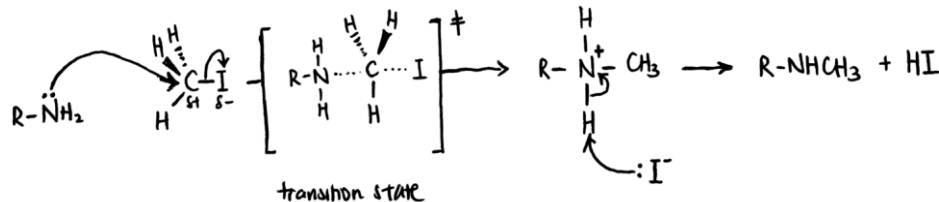
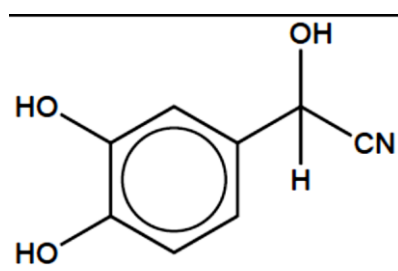
	<u>(B) Preparation of Reaction Mixture 1</u> 1. Weigh accurately about 6.0 g of propanoic acid into a clean and dry 100 cm³ conical flask. 2. Weigh accurately about 5.0 g of methanol in a weighing bottle and transfer to the same conical flask. Reweigh the weighing bottle to determine the actual mass of methanol used. 3. Using a 10.0 cm³ pipette, pipette 10.0 cm³ of HCl into the conical flask. 4. Stopper the conical flask with a rubber bung. Swirl the conical flask to mix the contents. 5. Let the reaction mixture stand for 1 week. <u>(B) Preparation of Reaction Mixture 2</u> 1. Repeat the experiment above using 4.0 g of propanoic acid. All other quantities remain the same. <u>Titration</u> 1. After 1 week, remove the stopper from the conical flask and add 2-3 drops of phenolphthalein into the conical flask. 2. Titrate the acids in the reaction mixture with 1.00 mol dm⁻³ NaOH(aq) (in a burette). 3. Observe the end point colour change when colour changes from colourless to pale pink, adding NaOH(aq) dropwise near the end point. <u>(C) Determining initial and equilibrium amounts of reactants and products</u> <u>For reaction mixture 1,</u> Initial $n_{\text{propanoic acid}} = \frac{6.0}{74.0} = 8.11 \times 10^{-2} \text{ mol}$ Initial $n_{\text{methanol}} = \frac{5.0}{32.0} = 1.56 \times 10^{-2} \text{ mol}$ Initial $n_{\text{HCl}} = \frac{10.0}{1000} \times 3 = 3.00 \times 10^{-2} \text{ mol (approx)}$ Initial $n_{\text{water}} = \frac{10}{18.0} = 0.556 \text{ mol}$ For the reversible reaction, $\text{CH}_3\text{CH}_2\text{CO}_2\text{H} + \text{CH}_3\text{OH} \rightleftharpoons \text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_3 + \text{H}_2\text{O}$ Let the average titre value of the titration with the reaction mixture be V_2. $n_{\text{NaOH required}} = \frac{V_2}{1000} \times 1.00 = (1.00 \times 10^{-3})V_2 \text{ mol}$ $= n_{\text{CH}_3\text{CH}_2\text{CO}_2\text{H}} + n_{\text{HCl present at equilibrium}}$ $n_{\text{CH}_3\text{CH}_2\text{CO}_2\text{H}} + n_{\text{HCl present initially}} = 8.11 \times 10^{-2} + 3.00 \times 10^{-2} \text{ mol} = 0.111 \text{ mol}$ $n_{\text{propanoic acid reacted}} = 0.111 - (1.00 \times 10^{-3})V_2 \text{ mol}$	
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If we let the amount of propanoic acid reacted be x, the equilibrium amounts of $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$, CH_3OH, $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_3$ and H_2O can then be calculated. The amount of HCl catalyst will remain constant as it is constantly regenerated.					
	$\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$	CH_3OH	H^+	$\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_3$	H_2O
Initial amt /mol	8.11×10^{-2}	1.56×10^{-2}	3.00×10^{-2}	0	0.556
Change/ mol	-x	-x	0 (catalyst)	+x	+x
Eqm amt /mol	$8.11 \times 10^{-2} - x$	$1.56 \times 10^{-2} - x$	3.00×10^{-2}	x	$0.556 + x$
$\text{CH}_3\text{CH}_2\text{CO}_2\text{H} + \text{CH}_3\text{OH} \rightleftharpoons \text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_3 + \text{H}_2\text{O}$ Since the stoichiometric coefficients of the reactants is equal to the stoichiometric coefficients of the products, $K_c = \frac{[\text{ester}][\text{water}]}{[\text{carboxylic acid}][\text{alcohol}]} = \frac{n_{\text{ester}}n_{\text{water}}}{n_{\text{carboxylic acid}}n_{\text{alcohol}}}$ <u>Safety Precaution</u> 1. Use gloves to handle the organic chemicals like propanoic acid to avoid direct contact with corrosive chemicals 2. Do not carry out any the experiment near a naked flame due to flammable compounds like methanol. 3. Conduct the experiment in the fumehood to prevent exposure to toxic fumes of methanol.					

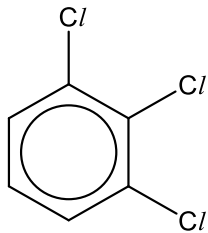
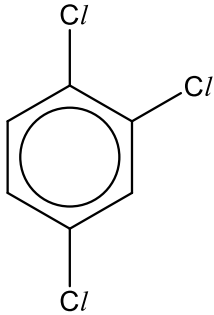
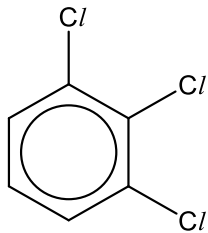
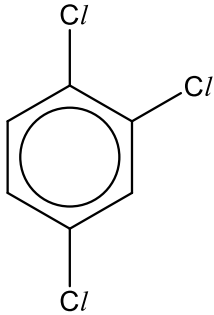
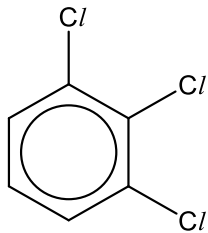
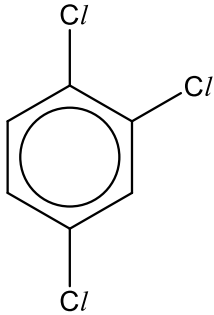
2	(a)	(i)	It is the enthalpy change when 1 mole of substance is completely burnt in excess of oxygen under standard conditions (298K, 1 bar).	[2]
		(ii)	Petrol is a mixture of compounds and comprise varying amounts of different compounds.	[1]
	(b)	(i)	Energy per $\text{dm}^3 = 0.0358 \times \frac{24.8 \times 10^6}{101000} = 8.79 \text{ MJ}$ [Students should use the data in the table given to solve the question]	[1]
		(ii)	Hydrogen produces water as the only by-product, while methane produces carbon dioxide, which is a greenhouse gas/contributes to global warming. Methane is also a greenhouse gas.	[1]
	(c)		There are hydrogen bonds that can be formed between methanol molecules. In addition, methanol has a higher number of electrons than methane and will have stronger instantaneous dipole-induced dipole interactions between molecules. These stronger intermolecular forces results in a higher boiling point for methanol.	[2]
	(d)	(i)	$\Delta H^\ominus = -726 - (-286 \times 2) - (-283) = +129 \text{ kJ mol}^{-1}$	[2]
		(ii)	$\Delta G^\ominus = \Delta H^\ominus - T\Delta S^\ominus = +129 - T(0.332)$ For ΔG to be negative, the minimum temperature is $\frac{129}{0.332} = 389 \text{ K}$	[2]
	(e)		Both O_2 and CO can act as <u>ligands</u> with the Fe^{2+} in haem group. CO , a stronger ligand, forms a <u>stronger dative covalent bond</u> with Fe^{2+} in haem. The CO molecule is irreversibly adsorbed at the site and this will destroy the haemoglobin oxygen carrying capacity.	[2]

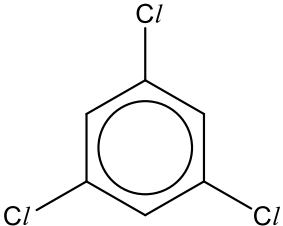
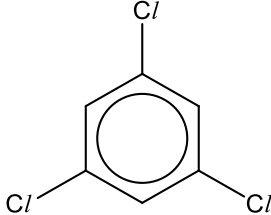
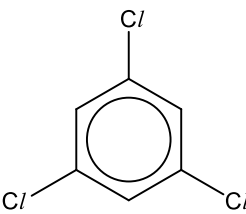
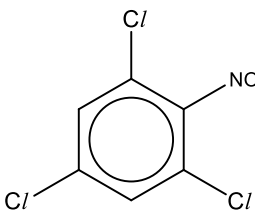
3	(a)	(i) <u>Bonds broken</u> <u>Bonds formed</u> 1 × H–H 2 × H–Cl 1 × Cl–Cl $\Delta H_r = [(+436) + (+244)] - [2(+431)]$ $= -182 \text{ kJ mol}^{-1}$	[1]
		(ii) 	[3]
	(b)	(i) Energy from the bright light is needed to start the reaction by forming chlorine radicals.	[1]
		(ii) Free radical substitution. $\text{Cl}_2 \xrightarrow{\text{uv}} 2\text{Cl}\bullet$	[2]
		(iii) The brief exposure to light is necessary to homolytically cleave the Cl–Cl bond to kickstart the free radical substitution mechanism. Since the reaction is exothermic, it would result in an increase in temperature, which increases the rate of reaction.	[2]
	(c)	(H–H bond energy = +436 kJ mol⁻¹; Cl–Cl bond energy = +244 kJ mol⁻¹ Br–Br bond energy = +193 kJ mol⁻¹; H–Cl bond energy = +431 kJ mol⁻¹ H–Br bond energy = +366 kJ mol⁻¹) $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{HCl}(\text{g}) \quad \Delta H_1$ $\text{H}_2(\text{g}) + \text{Br}_2(\text{g}) \rightarrow 2\text{HBr}(\text{g}) \quad \Delta H_2$ $\Delta H_1 = -182 \text{ kJ mol}^{-1}$ (from (a)(i)) $\Delta H_2 = (436 + 193) - 2(366) = -103 \text{ kJ mol}^{-1}$ Reaction to form the hydrogen halide involves breaking the Cl–Cl and Br–Br bonds, and forming H–Cl and H–Br bonds. Reaction 1 is more exothermic than reaction 2 as the H–Cl bond formed in reaction 1 is much stronger than the H–Br bond formed in reaction 2. Hence,	[3]

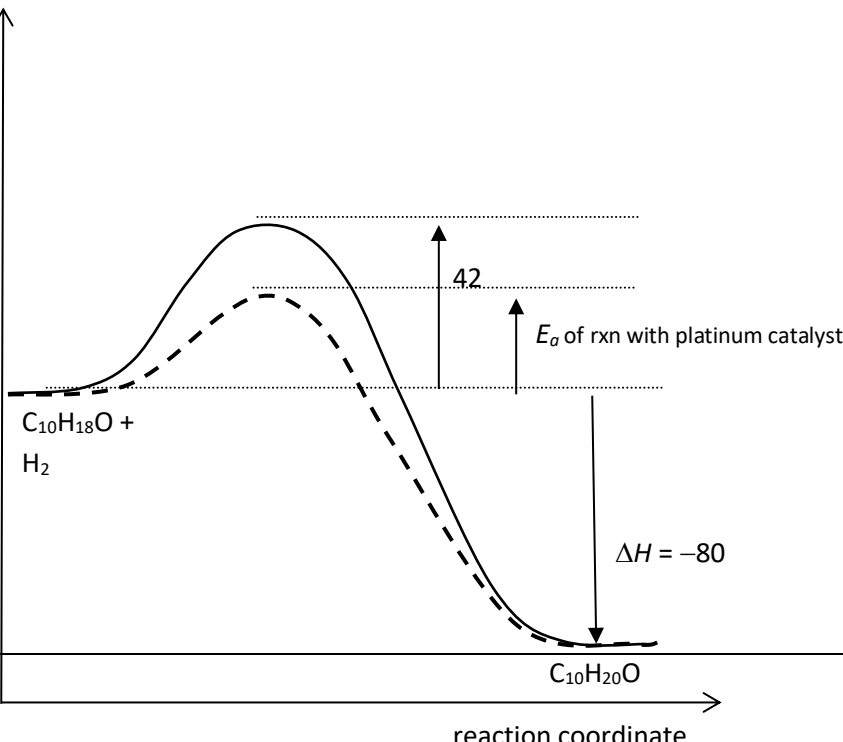
		reaction 1 proceeds towards completion explosively while reaction 2 proceeds at a lower rate.	
(d)	(i)	(Not in 9729 syllabus) $\text{NaCl(s)} + \text{H}_2\text{SO}_4(\text{l}) \rightarrow \text{NaHSO}_4(\text{s}) + \text{HCl(g)}$	[1]
	(ii)	(Not in 9729 syllabus) $2\text{HBr(g)} + \text{H}_2\text{SO}_4(\text{l}) \rightarrow \text{SO}_2(\text{g}) + \text{Br}_2(\text{g}) + 2\text{H}_2\text{O(l)}$ HBr is a stronger reducing agent than HCl and can reduce H_2SO_4 to SO_2 while it is oxidised to Br_2 , whereas HCl cannot reduce H_2SO_4 . Thus, yield of HBr will be low for this method.	[2]
(e)	1. Add aqueous chlorine dropwise to a small volume of each solution. Observation for HCl: Solution remain colourless. Observation for HBr: Solution turn from colourless to orange. (Reason: Displacement reaction happened between Cl_2 and HBr. $\text{Cl}_2 + 2\text{Br}^- \rightarrow \text{Br}_2 + 2\text{Cl}^-$) 2. Add acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ to a small volume of each solution and heat. Observation for HCl : solution remains orange Observation for HBr: solution turns orange to green (Reason: $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightleftharpoons 2\text{Cr}^{3+} + 7\text{H}_2\text{O} \quad E^\ominus = +1.33 \text{ V}$ $\text{Cl}_2 + 2\text{e}^- \rightleftharpoons 2\text{Cl}^- \quad E^\ominus = +1.36 \text{ V} \quad E^\ominus_{\text{cell}} = 1.33 - 1.36 = -0.03\text{V}$ $\text{Br}_2 + 2\text{e}^- \rightleftharpoons 2\text{Br}^- \quad E^\ominus = +1.07 \text{ V} \quad E^\ominus_{\text{cell}} = 1.33 - 1.07 = 0.26\text{V}$ Reaction is energetically feasible for HBr)		[3]

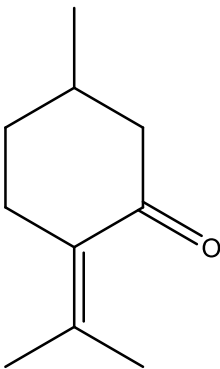
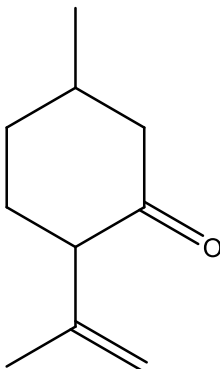
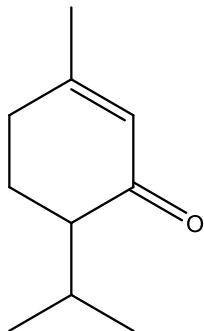
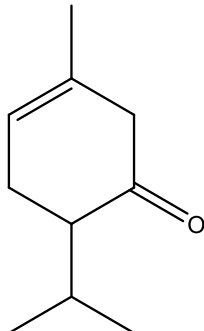
4	(a)		[2]
	(b) (i)		[4]
	(ii)	Iodomethane is a primary alkyl halide. The hydrogen atoms around the central carbon atom are small and hence there is little steric hindrance for the attack of the nucleophile (amine group of adrenaline) on the central carbon. OR The methyl carbocation (CH_3^+) formed through an $\text{S}_{\text{N}}1$ mechanism will not be stable due to the lack of electron-donating alkyl groups around the carbon atom.	[2]
	(c) (i)	 step 1: HCN, trace amount of NaCN/NaOH step 2: LiAlH_4 in dry ether	[3]
	(ii)	A racemic mixture, formed in step 1, contains both enantiomers of adrenaline. Only one of the enantiomers has the correct 3-dimensional shape to fit the specific geometry of the protein receptors in the human body.	[2]
	(d) (i)	$\text{Ag}_2\text{O} + 2\text{H}^+ + 2\text{e}^- \rightarrow 2\text{Ag} + \text{H}_2\text{O}$	[1]

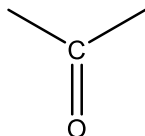
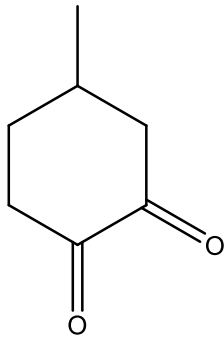
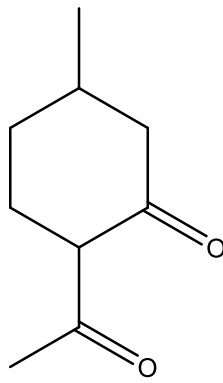
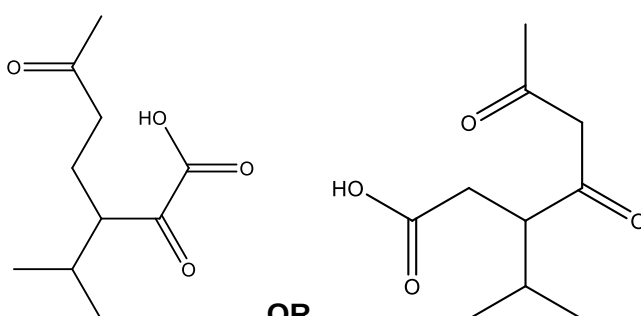
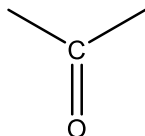
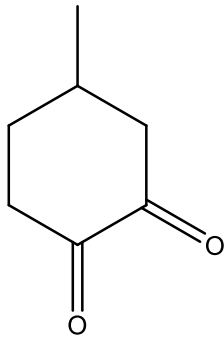
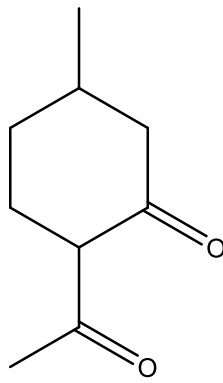
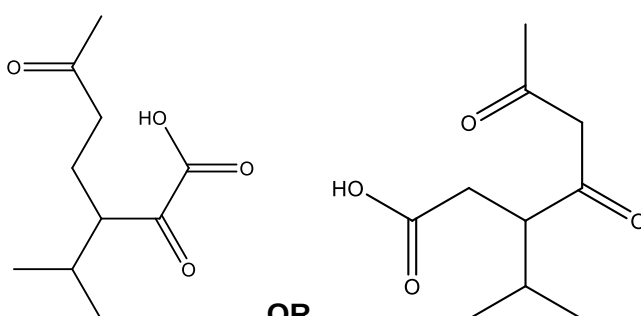
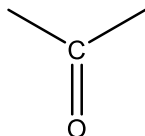
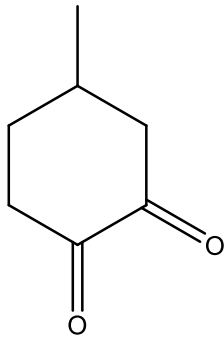
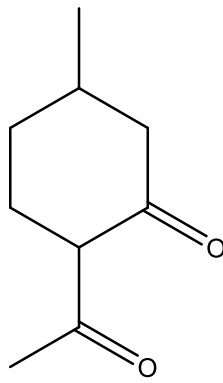
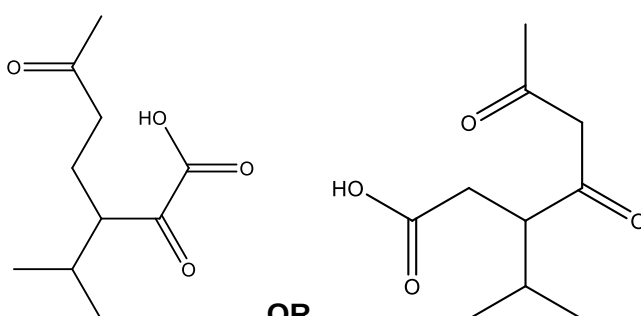
		(ii) $\text{C}_9\text{H}_{13}\text{NO}_3 + 2\text{Ag}_2\text{O} \rightarrow \text{C}_9\text{H}_9\text{NO}_3 + 4\text{Ag} + 2\text{H}_2\text{O}$ Amount of $\text{C}_9\text{H}_{13}\text{NO}_3 = \frac{50}{1000} \times 3.0 \times 10^{-3} = 1.50 \times 10^{-4} \text{ mol}$ Amount of $\text{Ag}_2\text{O} = 2 \times 1.50 \times 10^{-4} = 3.00 \times 10^{-4} \text{ mol}$ Mass of $\text{Ag}_2\text{O} = 3.00 \times 10^{-4} \times (2 \times 108 + 16.0) = \underline{\underline{0.0696 \text{ g}}}$	[2]
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5	(a)	Compounds X and Y are aryl halides. OR They have the halogen atoms directly bonded to the benzene ring. C–Cl bonds in these compounds have partial double bond character due to the delocalisation of a lone pair of electrons on the chlorine atom into the benzene ring. Hence, C–Cl bonds are relatively stronger and the compounds will not undergo nucleophilic substitution with NaOH(aq).				[2]																
	(b)	(i)	<table><tr><td></td><td>C</td><td>H</td><td>Cl</td></tr><tr><td>mass ratio</td><td>49</td><td>2.7</td><td>48.3</td></tr><tr><td>mole ratio</td><td>$\frac{49}{12.0} = 4.08$</td><td>$\frac{2.7}{1.0} = 2.7$</td><td>$\frac{48.3}{35.5} = 1.36$</td></tr><tr><td>simplest ratio</td><td>$\frac{4.08}{1.36} = 3$</td><td>$\frac{2.7}{1.36} = 1.98 \approx 2$</td><td>$\frac{1.36}{1.36} = 1$</td></tr></table> Empirical formula: <u>C₃H₂Cl</u>		C	H	Cl	mass ratio	49	2.7	48.3	mole ratio	$\frac{49}{12.0} = 4.08$	$\frac{2.7}{1.0} = 2.7$	$\frac{48.3}{35.5} = 1.36$	simplest ratio	$\frac{4.08}{1.36} = 3$	$\frac{2.7}{1.36} = 1.98 \approx 2$	$\frac{1.36}{1.36} = 1$			[2]
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		(ii)	pV = nRT = (m/M)RT Molar Mass (M) = $\frac{mRT}{pV} = \frac{0.344(8.31)(181+273)}{(101 \times 10^3)(87.4 \times 10^{-6})} = 147 \text{ g mol}^{-1}$ Let molecular formula of X be (C ₃ H ₂ Cl) _n [3(12.0) + 2(1.0) + 35.5] × n = 147 n = 2 Molecular formula: <u>C₆H₄Cl₂</u>			[4]																
	(c)	<table><tr><th>Structure</th><th>Name</th></tr><tr><td></td><td>1,2,3-trichlorobenzene</td></tr><tr><td></td><td>1,2,4-trichlorobenzene</td></tr></table>		Structure	Name		1,2,3-trichlorobenzene		1,2,4-trichlorobenzene			[3]										
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			1,3,5-trichlorobenzene	
	(d)	Structure of Y:   $+ \text{HNO}_3(\text{concentrated}) \xrightarrow[\text{maintain at } > 55\text{ }^\circ\text{C}]{\text{conc. H}_2\text{SO}_4}$  $+ \text{H}_2\text{O}$		[2]

1	(a)	(i)	Mol ratio of menthol : H_2 = 2:1 Amount of menthol = $1.32 \times 10^{-2} \times 2 = 0.0264$ mol Mass of menthol = $0.0264 \times (12.0 \times 10 + 20 \times 1.0 + 16.0) = 4.118$g % mass of menthol in peppermint oil = $\frac{4.118}{10.0} \times 100\% = 41.2\%$	[2]
		(ii)	3 chiral centres. No. of optical isomers : $2^3 = 8$	[2]
	(b)	(i) (ii)	Legend : ——— reaction pathway for Ni catalysed- reduction - - - - - reaction pathway for Pt catalysed-reduction  Energy /kJ mol⁻¹ 42 E_a of rxn with platinum catalyst $\Delta H = -80$ $C_{10}H_{18}O + H_2$ $C_{10}H_{20}O$ reaction coordinate	[2] + [1]
		(iii)	Nickel acts as a heterogeneous catalyst. The mode of action generally involves three steps: Adsorption: Menthone and hydrogen gas molecules diffuse to the surface of the Ni catalyst and form weak interactions (van der Waals' forces) to the active sites on the Ni catalyst surface. This weakens or breaks the bonds within the molecules themselves, making them more reactive. Reaction: The menthone and hydrogen molecules are held on the catalyst surface in close proximity and the correct orientation so that they can readily react together (thus lowering the activation energy of the reaction).	[2]

			3. Desorption: Weak interactions between menthol molecules and Ni catalyst break and the molecules diffuse away from the surface of the Ni catalyst and the active sites become available again.	
		(iv)	Iron/Palladium (or other transition elements) can be used as it is a transition element with partially filled 3d subshell.	[1]
(c)	<u>C:H ratio</u> in A, B, C $\approx$ 1:2		<u>Cycloalkane</u> present	
	A, B, C undergoes <u>electrophilic addition</u> with bromine water		C=C bond is present in A, B, C	
	A, B, C undergoes <u>condensation</u> with 2,4 DNPH		Aldehyde / ketone is present in A, B, C	
	None undergoes <u>oxidation</u> with Fehling's reagent		A, B, C has <u>ketone</u> functional group	
	A, B, C undergoes <u>reduction</u> with H ₂ /Pt catalyst.			
	A, B, C undergoes <u>oxidation</u> with acidified KMnO ₄ .			
	D, F, G give a positive iodoform test with alkaline aqueous iodine		Methyl ketone is present in D, F, G	
	   OR 			
				[10]

		<table><tr><td>A</td><td>B</td><td>C</td></tr><tr><td></td><td></td><td></td></tr><tr><td>D</td><td>E</td><td>F</td></tr><tr><td colspan="3"> OR (must correspond to C) </td></tr><tr><td colspan="3">G</td></tr></table>	A	B	C				D	E	F	 OR (must correspond to C)			G			
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G																		
2	(a)	(Volatility is the tendency of a substance to vapourise and depends on the boiling point of the substance. A compound with a <u>higher</u> boiling point would be less volatile.) Halogens exist as discrete molecules held together by weak van der Waals' forces (instantaneous dipole-induced dipole interactions) in a simple covalent structure. Down the group, the <u>number of electrons increases</u> (the electron cloud of each molecule increases in size and becomes more polarisable), hence <u>more energy is required to overcome the stronger van der Waals' forces between the halogen molecules</u>. This results in an <u>increase in boiling point and a decrease in volatility</u> down the group.	[2]															
	(b)	(Not in 9729 syllabus) $3\text{Cl}_2(\text{g}) + 6\text{OH}^-(\text{aq}) \rightarrow 5\text{Cl}^-(\text{aq}) + \text{ClO}_3^-(\text{aq}) + 3\text{H}_2\text{O}(\text{l})$ Oxidation state of Cl changed from 0 in Cl_2 to -1 in Cl^- and $+5$ in ClO_3^-.	[2]															
	(c)	<table><tr><td>(i)</td><td> H–F bond energy = $+562 \text{ kJ mol}^{-1}$ A large amount of energy is required to overcome the strong bond so as to release H^+ from HF. </td></tr><tr><td>(ii)</td><td> For HCl , </td></tr></table>	(i)	H–F bond energy = $+562 \text{ kJ mol}^{-1}$ A large amount of energy is required to overcome the strong bond so as to release H^+ from HF.	(ii)	For HCl ,	<table><tr><td>[1]</td></tr><tr><td>[2]</td></tr></table>	[1]	[2]									
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		pH = -lg (0.50) = <u>0.301</u> For HF, <table><tr><td></td><td>HF(aq) ⇌ F⁻(aq) + H⁺(aq)</td><td></td><td></td></tr><tr><td>Initial concentration/ mol dm⁻³</td><td>0.50</td><td>0</td><td>0</td></tr><tr><td>Change in concentration/ mol dm⁻³</td><td>-x</td><td>+x</td><td>+x</td></tr><tr><td>Equilibrium concentration/ mol dm⁻³</td><td>0.50 - x</td><td>x</td><td>x</td></tr></table> $K_a = [\text{F}^-][\text{H}^+] / [\text{HF}] = x^2 / (0.50 - x)$ Assume 0.50 - x ≈ 0.50, $K_a = 5.6 \times 10^{-4} \approx x^2 / 0.50$ $[\text{H}^+] = x = 0.0167 \text{ mol dm}^{-3}$ pH = -lg (0.0167) = <u>1.78</u>		HF(aq) ⇌ F ⁻ (aq) + H ⁺ (aq)			Initial concentration/ mol dm ⁻³	0.50	0	0	Change in concentration/ mol dm ⁻³	-x	+x	+x	Equilibrium concentration/ mol dm ⁻³	0.50 - x	x	x	
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	(d)	When silver nitrate is added to aqueous chloride ion, a white ppt AgCl is formed. When aqueous ammonia is added, silver chloride dissolves to form a colourless solution due to the formation of the diamminesilver(I) complex ion, [Ag(NH₃)₂]⁺(aq) occur. $\text{Ag}^+(\text{aq}) + 2\text{NH}_3(\text{aq}) \rightarrow [\text{Ag}(\text{NH}_3)_2]^+(\text{aq})$ When silver nitrate is added to aqueous iodide ions, a yellow ppt AgI is formed. The yellow ppt is insoluble in aqueous ammonia.	[3]																
	(e)	(i)	At constant temperature, <i>pV</i> is a constant.	[1]															
		(ii)	$V = (0.40)(8.31)(300) / (12.0 \times 10^5) = 8.31 \times 10^{-4} \text{ m}^3 = 0.831 \text{ dm}^3$	[1]															
		(iii)	<table><tr><td>Pressure/Pa</td><td>Volume/dm³</td><td><i>pV</i> / Pa dm³</td></tr><tr><td>5.0 × 10⁵</td><td>1.924</td><td>9.62 × 10⁵</td></tr><tr><td>10.0 × 10⁵</td><td>0.926</td><td>9.26 × 10⁵</td></tr><tr><td>15.0 × 10⁵</td><td>0.592</td><td>8.88 × 10⁵</td></tr><tr><td>12.0 × 10⁵</td><td>0.753</td><td>9.11 × 10⁵</td></tr></table> <i>pV</i> at 12.0 × 10⁵ Pa = (9.26 × 10⁵ – 8.88 × 10⁵) × 3/5 + 8.88 × 10⁵ = 9.11 × 10⁵ Pa dm³ $V = (9.11 \times 10^5) / (12.0 \times 10^5) = 0.759 \text{ dm}^3$	Pressure/Pa	Volume/dm ³	<i>pV</i> / Pa dm ³	5.0 × 10 ⁵	1.924	9.62 × 10 ⁵	10.0 × 10 ⁵	0.926	9.26 × 10⁵	15.0 × 10 ⁵	0.592	8.88 × 10⁵	12.0 × 10 ⁵	0.753	9.11 × 10⁵	[2]
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		(iv)	The answer in part (ii) is based on ideal gas law <i>pV</i> = <i>nRT</i> . The law has the assumption that the intermolecular forces between the gaseous molecules are negligible. However, HCl is a polar molecule, and permanent dipole-permanent dipole interactions exist between HCl molecules. This results in actual volume (iii) HCl being lower than ideal gas law calculations (ii).	[2]															
	(f)	(i)	$\Delta G = \Delta H - T\Delta S$ $\Delta S = (\Delta H - \Delta G) / T = (+16.8 - 0.0)/188 = +0.0894 \text{ kJ mol}^{-1} \text{ K}^{-1}$	[2]															

			Since ΔS is positive, there is an increase in disorder due to an increase in number of gaseous molecules.	
		(ii)	$\Delta G = \Delta H - T\Delta S = +16.8 - (298)(0.0894) = -9.83 \text{ kJ mol}^{-1}$ ΔG becomes negative, the reaction is spontaneous at higher temperatures.	[2]

3	(a)	(i)	(Not in 9729 syllabus) Magnesium burns in oxygen with a brilliant white flame to form a white solid (MgO). Calcium burns in oxygen with a brick-red flame to form a white solid (CaO).	[2]
		(ii)	MgO(s) <u>reacts</u> with water to form Mg(OH)₂(s), which <u>dissolves sparingly</u> in water to give a <u>weakly alkaline</u> (pH = 8) solution. $\text{MgO(s)} + \text{H}_2\text{O(l)} \rightarrow \text{Mg(OH)}_2\text{(s)}$ $\text{Mg(OH)}_2\text{(s)} \rightleftharpoons \text{Mg}^{2+}\text{(aq)} + 2\text{OH}^-\text{(aq)}$. (CaO not in 9729 syllabus) CaO dissolves in water to form a colourless alkaline solution of Ca(OH)₂.	[1]
	(b)	(i)	Any value between the solubility of Ca(OH) ₂ and Ba(OH) ₂ , e.g. $8.0 \times 10^{-2} \text{ mol dm}^{-3}$	[1]
		(ii)	$[\text{OH}^-] = 2(2.5 \times 10^{-2}) = 5.0 \times 10^{-2} \text{ mol dm}^{-3}$ $\text{pOH} = -\lg(5.0 \times 10^{-2}) = 1.30$ $\text{pH} = 14 - 1.30 = \underline{12.7}$	[2]

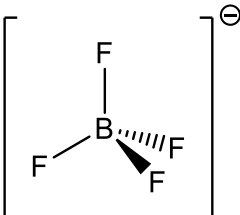
		(iii)	$K_{sp} = [\text{Mg}^{2+}][\text{OH}^-]^2$ $= (1.6 \times 10^{-4})(2 \times 1.6 \times 10^{-4})^2$ $= \underline{1.64 \times 10^{-11} \text{ mol}^3 \text{ dm}^{-9}}$	[2]				
		(iv)	A white precipitate of $\text{Mg}(\text{OH})_2$ will be formed. OH^- , is a common ion and from both salt solutions. Since K_{sp} of $\text{Mg}(\text{OH})_2 < K_{sp}$ of $\text{Ba}(\text{OH})_2$, the concentration of OH^- in the resulting solution is higher than that of $[\text{OH}^-]$ in saturated $\text{Mg}(\text{OH})_2$ and lower than of $[\text{OH}^-]$ in saturated $\text{Ba}(\text{OH})_2$. Thus the K_{sp} of $\text{Mg}(\text{OH})_2$ will be exceeded and some $\text{Mg}(\text{OH})_2$ will precipitate out.	[2]				
	(c)	(i)	(Not in 9729 syllabus) aspartate and glutamate. Type of bonding: ionic interactions	[2]				
		(ii)	(Not in 9729 syllabus) α -helix is a regular spiral configuration of the polypeptide chain, held in place by hydrogen bonds between the backbone C=O group of one amino acid residue and the backbone N-H group of the 4th amino acid residue further down the chain.	[2]				
		(iii)	$\text{H}_2\text{SO}_4/\text{HCl}(\text{aq})$, heat (for several hours) OR $\text{NaOH}/\text{KOH}(\text{aq})$, heat (for several hours)	[1]				
		(iv)	(Not in 9729 syllabus) gly-asp-gly-tyr-ile-ser	[1]				
	(d)	(i)	<table><tr><td></td><td></td></tr><tr><td>H</td><td>J</td></tr></table>			H	J	[2]
H	J							
		(ii)	Step 4: (Excess) Ethanolic NH_3 , heat in sealed tube Step 5: Dilute $\text{H}_2\text{SO}_4/\text{HCl}(\text{aq})$, heat	[2]				

4	(a)	step 1: hydrolysis step 2: elimination (of water) step 3: reduction	[3]						
	(b)	$\text{C}_6\text{H}_8\text{O(l)} + \frac{15}{2} \text{O}_2\text{(g)} \rightarrow 6\text{CO}_2\text{(g)} + 4\text{H}_2\text{O(l)}$<table><tr><th colspan="2">Bonds broken</th><th>Bonds formed</th></tr><tr><td>3 C–C</td><td>2 C=C</td><td>12 C=O</td></tr></table>	Bonds broken		Bonds formed	3 C–C	2 C=C	12 C=O	[3]
Bonds broken		Bonds formed							
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		$\begin{array}{ccc} 2 \text{ C-O} & 8 \text{ C-H} & 8 \text{ O-H} \\ \\ \frac{15}{2} \text{ O=O} & & \end{array}$ $\Delta H = E(\text{bonds broken}) - E(\text{bonds formed})$ $= [(3 \times 350) + (2 \times 610) + (2 \times 360) + (8 \times 410) + (\frac{15}{2} \times 496)] - [(12 \times 805) + (8 \times 460)]$ $= \underline{-3350 \text{ kJ mol}^{-1}}$	
	(c)	Heat absorbed by water = $(200)(4.18)(32) = 26752 \text{ J}$ Heat evolved from combustion of DMF = $\frac{26752}{80} \times 100 = 33440 \text{ J} = 33.44 \text{ kJ}$ Amount of DMF = $\frac{1.00}{96} = 0.0104 \text{ mol}$ $\Delta H = -\frac{33.44}{0.0104} = \underline{-3.21 \times 10^3 \text{ kJ mol}^{-1}}$ The bond energy value in the Data Booklet represents the <u>average bond dissociation energy</u> (of that particular bond in the full range of molecules that contain the bond, whereas the experimental value involves forming and breaking specific bonds in the given reaction). (Alternatively, the bond energy values in the Data Booklet involve the breaking of covalent bonds in gaseous molecules, whereas the experiment involves combustion of liquid DMF.)	[4]
	(d)	(i) Step 1: $\text{R-CH}_2\text{OH} + \text{H}^+ \rightarrow \text{R-CH}_2\text{OH}_2^+$ fast Step 2: $\text{R-CH}_2\text{OH}_2^+ \rightarrow \text{R-CH}_2^+ + \text{H}_2\text{O}$ slow Step 3: $\text{R-CH}_2^+ + \text{Cl}^- \rightarrow \text{R-CH}_2\text{Cl}$ fast	[3]
		(ii) $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$, $\text{H}_2\text{SO}_4(\text{aq})$, heat under reflux (acidified KMnO_4 will oxidise the double bond, hence should not be used)	[1]
	(e)	(i) Ester functional group	[1]
		(ii) $\begin{array}{cc} \text{OH} & \text{OH} \\ & \\ \text{H}-\text{C} & -\text{C}-\text{H} \\ & \\ \text{H} & \text{H} \end{array}$	[1]
		(iii) (Ethane-1,2-diol and a few drops of) concentrated H_2SO_4, heat under reflux	[1]

	(f)	Free radical substitution <u>Initiation</u> $\text{Cl}_2 \xrightarrow[\text{light}]{\text{uv}} 2\text{Cl}\bullet$ <u>Propagation</u> $\text{R-CH}_3 + \text{Cl}\bullet \rightarrow \text{R-CH}_2\bullet + \text{HCl}$ $\text{R-CH}_2\bullet + \text{Cl}_2 \rightarrow \text{R-CH}_2\text{Cl} + \text{Cl}\bullet$ <u>Termination</u> $2\text{Cl}\bullet \rightarrow \text{Cl}_2$ $\text{R-CH}_2\bullet + \text{R-CH}_2\bullet \rightarrow \text{R-CH}_2\text{CH}_2\text{-R}$ $\text{R-CH}_2\bullet + \text{Cl}\bullet \rightarrow \text{R-CH}_2\text{Cl}$	[3]
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5	(a)	(i)	Proton number refers to the number of protons in the nucleus of an atom. Nucleon number refers to the sum of the number of protons and neutrons in the nucleus of an atom.	[1]
		(ii)	Let % abundance of ${}^6\text{Li}$ be x. $\frac{x}{100}(6.015) + \left(\frac{100-x}{100}\right)(7.016) = 6.942$ $x = 7.39\%$ Abundance of ${}^6\text{Li} = \underline{7.39\%}$ (2 d.p.) Abundance of ${}^7\text{Li} = 100 - 7.39 = \underline{92.61\%}$ (2 d.p.)	[2]
		(iii)	X: ${}^3\text{He}$ Y: ${}^7\text{Li}^-$	[2]
	(b)	(i)	Metallic bonding. The valence electrons of Li atoms will delocalise together with the delocalised electrons between the layers of carbon atoms in graphite. Electrostatic forces of attraction will form between the Li^+ cations and delocalised electrons between the layers of carbon atoms.	[1]
		(ii)	Oxidation number of Co before discharge: +4 Oxidation number of Co after the cell is totally discharged: +3	[1]
		(iii)	Tetrahedral.	[1]

		There are 4 bond pairs and no lone pairs around the boron atom in BF_4^- . 	
	(iv)	Cold, dilute, alkaline $\text{KMnO}_4(\text{aq})$	[1]
	(v)	Condensation (OR Nucleophilic (acyl) substitution)	[1]
(c)	(i)	$\text{Li}_2\text{O}_2 + \text{CO}_2 \rightarrow \text{Li}_2\text{CO}_3 + \frac{1}{2}\text{O}_2$	[1]
	(ii)	Li^+ is smaller than the other Group I metal ions, and has a high charge density (similar to Mg^{2+}). Hence, Li^+ polarises the carbon-oxygen bonds in large CO_3^{2-} ion to a large extent, weakening the bonds. The bonds will be broken when Li_2CO_3 is heated.	[2]
	(d) (i)	$\text{CH}_3\text{CH}_2\text{Br}$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$ OR $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ and $\text{CH}_3\text{CH}_2\text{CHO}$	[2]
	(ii)	CH_3Br and $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$ OR $\text{CH}_3\text{CH}_2\text{Br}$ and $\text{CH}_3\text{CH}_2\text{COCH}_3$	[2]
	(e)	Heat a small sample of P, Q and R with acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$. For P: $\text{K}_2\text{Cr}_2\text{O}_7$ solution will turn from orange to green. For Q: $\text{K}_2\text{Cr}_2\text{O}_7$ solution will turn from orange to green. For R: $\text{K}_2\text{Cr}_2\text{O}_7$ solution will remain orange. R is distinguished from P and Q. To the products of P & Q above, add 2,4-DNPH (Brady's Reagent). For Q: orange or yellow precipitate forms. For P: no orange or yellow precipitate will form. Note: To use $\text{Na}_2\text{CO}_3(\text{aq})$ to distinguish P from Q, the oxidised product from heating with acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ needs to be isolated.	[3]